1 Probability and the Logic of Statistical Inference

Imagine that you open a weather app and see a 30% chance of rain. You see the probability but you are wondering what might actually happen. Now consider a different situation. You know that it rained on 7 of the last 10 days, and you want to learn something about the usual weather here in Santa Cruz. That reversal from using a probability to anticipate data, to using data to learn about a probability, is the move from **probability theory** to **statistical inference**.

This chapter reviews the probability ideas that will be most important as we study statistical inference. The aim is not to rederive every formula. Instead, we will focus on what each object describes, how the pieces fit together, and why they will matter later in the course.

1.1 Probability begins with uncertainty

A **random experiment** is any repeatable process whose outcome is not known in advance. The word *experiment* is used broadly. Rolling a die is an experiment, but so is recording the time until the next bus arrives, selecting a student at random and asking how long they slept, or observing whether a customer renews a subscription.

The **sample space**, usually written Ω, is the collection of every possible outcome. Its form depends on what we choose to record.

- If we record the result of one coin toss, $\Omega = \{H, T\}$.
- If we record a student's commute time in minutes, a useful sample space might be $\Omega = [0, \infty)$.
- If we record the outcomes of two coin tosses in order, $\Omega = \{HH, HT, TH, TT\}$.

The sample space is not simply "everything that could happen in the universe." It is everything our experiment is designed to record. For two coin tosses, "the second toss falls onto the floor" is not part of the mathematical model unless we decide to include it.

An **event** is a collection of outcomes (i.e., a subset of the sample space). In the two-toss example, the event "exactly one head" is

$$A = \{HT, TH\}.$$

An event occurs when the observed outcome belongs to its set. This language lets us ask probability questions about features of outcomes rather than only about individual outcomes.

1.1.1 What a probability measure does

A **probability measure** P assigns a number to each event. The number $P(A)$ describes how likely event A is under the model. Probabilities obey three basic rules:

1. $0 \leq P(A) \leq 1$ for every event A.
2. $P(\Omega) = 1$ because some outcome in the sample space must occur.
3. If A and B cannot occur together, then $P(A \cup B) = P(A) + P(B)$.

The third rule applies directly only to **disjoint** events. For example, on one die roll, the events "roll a 1" and "roll a 2" cannot both happen, so their probabilities can simply be added. In contrast, "roll an even number" and "roll a number greater than 3" overlap at 4 and 6. Adding their probabilities without correcting for that overlap would count those outcomes twice.

Probabilities can be interpreted as long-run proportions. If a process is repeated many times under stable conditions, an event with probability 0.30 should occur on roughly 30% of repetitions. This does not mean it must occur exactly 3 times in every 10 trials. Randomness will typically create short-run variation.

```r
set.seed(132)

tosses <- sample(c(0, 1), size = 1000, replace = TRUE)
running_proportion <- cumsum(tosses) / seq_along(tosses)

plot(
  running_proportion,
  type = "l",
  xlab = "Number of tosses",
  ylab = "Proportion of heads",
  ylim = c(0, 1),
  col = "steelblue",
  lwd = 2
)
abline(h = 0.5, col = "firebrick", lty = 2, lwd = 2)
```

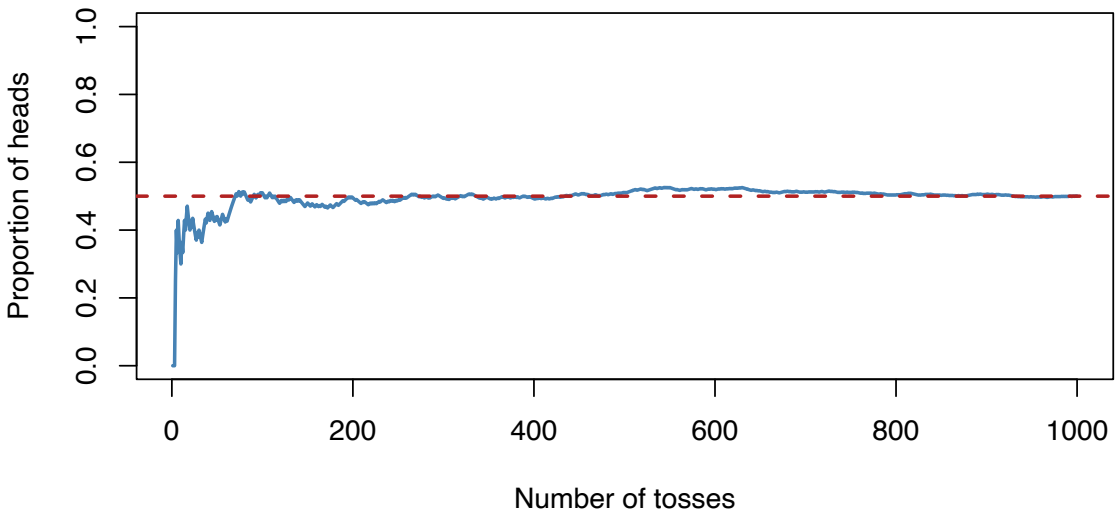


Figure 1.1: The proportion of heads fluctuates at first and then settles near 0.50.

In Figure 1.1, a head is coded as 1 and a tail as 0. After only a few tosses, one additional head can change the proportion substantially. After hundreds of tosses, one result has little effect, and the proportion tends to remain near 0.50.

1.2 Conditional probability

Often a probability changes when we learn something new. The **conditional probability** of A given B is

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)}, \qquad P(B) > 0.$$

The vertical bar is read as "given." The formula says that once we know B occurred, outcomes outside B are no longer relevant. We restrict attention to B and ask what fraction of its probability also belongs to A.

Suppose a campus survey finds that 40% of students commute, 25% commute and have an 8 a.m. class, and 50% have an 8 a.m. class. Let

- C be the event that a student commutes, and

- E be the event that a student has an 8 a.m. class.

Then

$$P(C \mid E) = \frac{P(C \cap E)}{P(E)} = \frac{0.25}{0.50} = 0.50.$$

Among students with an 8 a.m. class, half commute. Notice that $P(C \mid E) = 0.50$ differs from the unconditional probability $P(C) = 0.40$. Knowing about the class schedule changed the relevant group.

The order matters. In the same example,

$$P(E \mid C) = \frac{P(C \cap E)}{P(C)} = \frac{0.25}{0.40} = 0.625.$$

"The probability a commuter has an 8 a.m. class" is not the same question as "the probability an 8 a.m. student commutes." Confusing these two directions is a common source of incorrect reasoning.

1.2.1 Bayes' rule

Bayes' rule provides a way to reverse the direction of a conditional probability:

$$P(A \mid B) = \frac{P(B \mid A)P(A)}{P(B)}.$$

For an event A and its complement A^c, the denominator can be expanded as

$$P(B) = P(B \mid A)P(A) + P(B \mid A^c)P(A^c).$$

Each part has a useful interpretation:

- $P(A)$ describes how plausible A was before observing B.
- $P(B \mid A)$ describes how compatible the observation B is with A.
- $P(A \mid B)$ describes how plausible A is after observing B.

Consider an automated plagiarism-screening tool. Suppose 2% of submitted papers actually contain copied passages. The tool flags 90% of papers that contain copied material, but it also flags 5% of papers that do not. If a paper is flagged, what is the probability that it actually contains copied material?

Let C mean that the paper contains copied material and F mean that it is flagged. Then

$$\begin{aligned} P(C \mid F) &= \frac{P(F \mid C)P(C)}{P(F \mid C)P(C) + P(F \mid C^c)P(C^c)} \\ &= \frac{(0.90)(0.02)}{(0.90)(0.02) + (0.05)(0.98)} \\ &\approx 0.269. \end{aligned}$$

Despite the tool's high detection rate, only about 27% of flagged papers actually contain copied material under these assumptions. The reason is the low starting rate. There are many more papers without copied material, so even a 5% false-positive rate creates many flags. This is why the **base rate** $P(C)$ matters.

Bayes' rule will later become the foundation of Bayesian inference, where data can be used to update what we believe about unknown parameters.

1.3 Random variables turn outcomes into numbers

A **random variable** is a function that assigns a numerical value to each outcome:

$$X : \Omega \longrightarrow \mathbb{R}.$$

The formal definition can sound more mysterious than the idea. A random variable is simply a numerical measurement determined by a random outcome.

For two coin tosses, define X as the number of heads. Then

$$X(HH) = 2, \qquad X(HT) = X(TH) = 1, \qquad X(TT) = 0.$$

The randomness comes from not knowing which outcome will occur. Once the experiment is performed, the observed value is an ordinary number. We typically use an uppercase letter such as X for the random variable and a lowercase letter such as x for a possible or observed value.

1.3.1 Discrete and continuous random variables

A **discrete random variable** takes values that can be listed, even if the list is infinite. Examples include the number of unread emails you have, the number of customers arriving in an hour, and the number of attempts needed to pass a driving test.

A **continuous random variable** can take any value in an interval. Examples include reaction time, temperature, distance, and battery life. A recorded measurement may be rounded (perhaps battery life is reported to the nearest minute) but the underlying quantity is still modeled as continuous.

The distinction matters because probabilities are assigned differently in the two cases.

1.4 Three ways to describe a distribution

A **probability distribution** describes which values a random variable can take and how probability is allocated among them (i.e., how likely are different values to occur). A distribution is often used as a model for random variation.

1.4.1 Probability mass functions

For a discrete random variable, the **probability mass function** (PMF) is

$$f(x) = P(X = x).$$

The PMF gives the probability of each possible value directly. All values are nonnegative and their probabilities sum to 1.

For the number of heads in two fair tosses,

x	0	1	2
$P(X = x)$	1/4	1/2	1/4

The middle value has more probability because two different outcomes, HT and TH, produce one head.

1.4.2 Probability density functions

For a continuous random variable, a **probability density function** (PDF) is a curve $f(x)$ for which probability is represented by area:

$$P(a \leq X \leq b) = \int_a^b f(x)\,dx.$$

The height of a density curve is not itself a probability. A tall part of the curve indicates that values are relatively concentrated nearby, but probability comes from the **area over an interval**. This distinction also explains why, for a continuous random variable,

$$P(X = x) = 0$$

for any single exact value x. An interval with zero width has zero area.

1.4.3 Cumulative distribution functions

The **cumulative distribution function** (CDF) works for both discrete and continuous random variables:

$$F(x) = P(X \leq x).$$

It accumulates all probability at or below some value x. A CDF never decreases, starts at 0 on the far left, and approaches 1 on the far right.

Suppose commute time is normally distributed with mean 25 minutes and standard deviation 6 minutes. In R, `pnorm()` evaluates the normal CDF:

```
mean_commute <- 25
sd_commute <- 6

# Probability that a commute takes at most 30 minutes
pnorm(30, mean = mean_commute, sd = sd_commute)
```

```
[1] 0.7976716
```

```
# Probability that a commute takes between 20 and 30 minutes
pnorm(30, mean = mean_commute, sd = sd_commute) -
  pnorm(20, mean = mean_commute, sd = sd_commute)
```

```
[1] 0.5953432
```

The first calculation collects all probability to the left of 30. The second subtracts the area to the left of 20 from the area to the left of 30, leaving only the area between them.

```r
commute_times <- seq(5, 45, length.out = 300)
density_values <- dnorm(commute_times, mean = 25, sd = 6)
cdf_values <- pnorm(commute_times, mean = 25, sd = 6)

par(mfrow = c(1, 2))

plot(
  commute_times, density_values,
  type = "l", lwd = 2, col = "steelblue",
  xlab = "Commute time (minutes)", ylab = "Density",
  main = "Probability density function"
)

plot(
  commute_times, cdf_values,
  type = "l", lwd = 2, col = "darkorange",
  xlab = "Commute time (minutes)", ylab = "Cumulative probability",
  main = "Cumulative distribution function"
)

par(mfrow = c(1, 1))
```

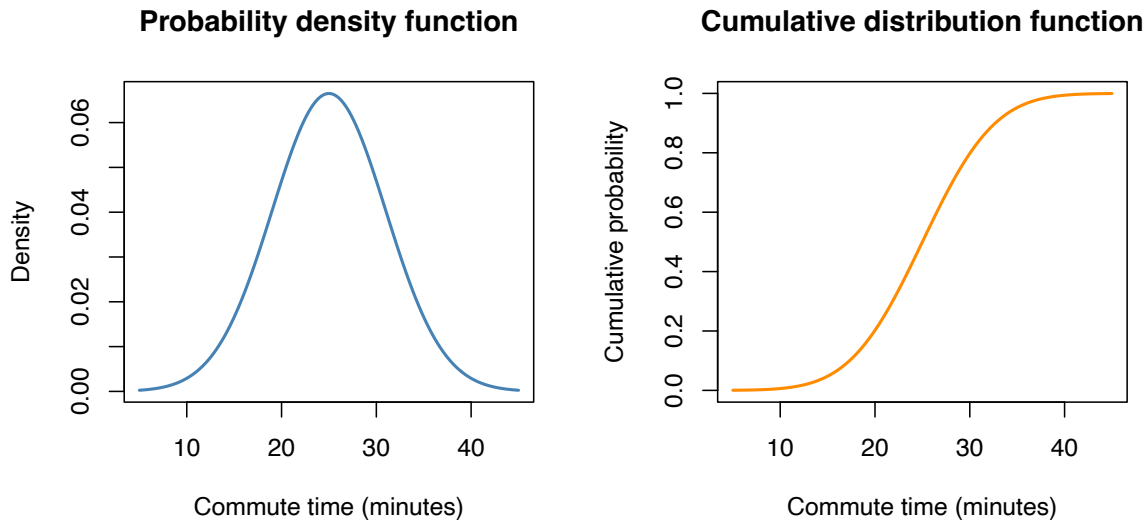


Figure 1.2: A density describes local concentration while a CDF shows accumulated probability.

In Figure 1.2, the density rises and falls, while the CDF can only rise. At 25 minutes, the CDF equals 0.5 meaning half of this distribution lies at or below its mean.

1.5 Expectation and variance summarize a distribution

A full distribution contains a great deal of information. Two numerical summaries (expectation and variance) can be used describe its center and spread.

1.5.1 Expected value

For a discrete random variable,

$$E[X] = \sum_x xP(X = x),$$

and for a continuous random variable,

$$E[X] = \int_{-\infty}^{\infty} xf(x)\,dx.$$

The **expected value**, or mean, is a probability-weighted average of the possible values. Values with more probability exert more influence on the mean.

Expectation is best understood as a long-run average, not a guaranteed result from one or a few trials. If a café gives a digital reward worth \$0, \$2, or \$10 with probabilities 0.70, 0.25, and 0.05, respectively, then

$$E[X] = (0)(0.70) + (2)(0.25) + (10)(0.05) = 1.$$

No customer receives exactly \$1, yet the average reward over many customers should be close to \$1. An expected value does not need to be one of the possible individual outcomes.

1.5.2 Variance and standard deviation

The **variance** of X is

$$\mathrm{Var}(X) = E\left[(X - E[X])^2\right] = E[X^2] - (E[X])^2.$$

Variance measures how far values tend to fall from the mean. The deviations are squared so that distances below and above the mean do not cancel. Squaring also gives greater weight to unusually large deviations.

Because variance is expressed in squared units, we often interpret the **standard deviation** instead:

$$\mathrm{SD}(X) = \sqrt{\mathrm{Var}(X)}.$$

If commute time is measured in minutes, its variance is measured in squared minutes, but its standard deviation is again measured in minutes.

Two courses could both have an average exam score of 75 while differing greatly in variability. In one, nearly everyone may score between 70 and 80. In the other, scores may range from 40 to 100. The same mean does not imply the same distribution.

1.6 Some common distributions

Probability distributions are reusable models for common forms of random variation. Choosing a distribution means making assumptions about the values that are possible and how their probabilities behave. Below are some commonly used distributions that describe frequently occuring patterns of variation.

1.6.1 Binomial

If $X \sim \text{Bin}(n, p)$, then

$$P(X = x) = \binom{n}{x} p^x (1-p)^{n-x}, \qquad x = 0, 1, \ldots, n.$$

The binomial distribution applies when there are n trials, each trial has two categorized outcomes, the success probability p stays constant, and the trials are independent. "Success" is only a label for the outcome being counted. It could mean a student answers correctly, a package arrives on time, or a free throw is made.

The term $\binom{n}{x}$ counts the number of different arrangements that contain x successes. The term $p^x(1-p)^{n-x}$ gives the probability of any one such arrangement.

Its mean and variance are

$$E[X] = np, \qquad \text{Var}(X) = np(1-p).$$

For example, if a person receives 20 legitimate emails and independently has a 10% chance of overlooking each one, the expected number overlooked is $20(0.10) = 2$.

1.6.2 Poisson

If $X \sim \text{Poisson}(\lambda)$, then

$$P(X = x) = \frac{\lambda^x e^{-\lambda}}{x!}, \qquad x = 0, 1, 2, \ldots .$$

The Poisson distribution often models the number of events occurring in a fixed interval of time or space. For example, customers arriving in the next ten minutes, potholes along a mile of road, or support requests received overnight. The parameter $\lambda > 0$ is the average number of events in that interval.

For a Poisson random variable,

$$E[X] = \lambda, \qquad \text{Var}(X) = \lambda.$$

That equality is a recognizable feature of the Poisson model. Real count data with variance much larger than the mean may need a different model.

1.6.3 Normal

If $X \sim N(\mu, \sigma^2)$, its density is

$$f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left[-\frac{(x-\mu)^2}{2\sigma^2}\right], \qquad -\infty < x < \infty.$$

The normal distribution is symmetric and bell-shaped. The parameter μ sets its center and σ^2 controls its spread:

$$E[X] = \mu, \qquad \mathrm{Var}(X) = \sigma^2.$$

It can be a useful model for measurements produced by many small sources of variation, such as manufacturing dimensions or repeated measurement error. It is not automatically appropriate for every continuous variable. Income, waiting time, and other strongly right-skewed variables are usually not well represented by a symmetric normal curve.

1.6.4 Exponential

If $X \sim \mathrm{Exp}(\lambda)$, then

$$f(x) = \lambda e^{-\lambda x}, \qquad x \geq 0.$$

The exponential distribution is often used for waiting times when events occur at a constant average rate. If requests reach a server at an average rate of 3 per minute, an exponential model may describe the time until the next request.

Its mean and variance are

$$E[X] = \frac{1}{\lambda}, \qquad \mathrm{Var}(X) = \frac{1}{\lambda^2}.$$

1.6.5 Gamma

One common rate-parameter form of the gamma density is

$$f(x) = \frac{\beta^\alpha}{\Gamma(\alpha)} x^{\alpha-1} e^{-\beta x}, \qquad x > 0,$$

where $\alpha > 0$ is the shape and $\beta > 0$ is the rate. Its mean and variance are

$$E[X] = \frac{\alpha}{\beta}, \qquad \mathrm{Var}(X) = \frac{\alpha}{\beta^2}.$$

The gamma distribution is flexible and right-skewed. It can model positive quantities such as service times or the time until several events have occurred. The exponential distribution is the special case $\alpha = 1$.

The gamma function in the denominator extends the factorial to positive real numbers:

$$\Gamma(x) = \int_0^\infty t^{x-1} e^{-t}\, dt, \qquad \Gamma(n) = (n-1)!$$

for a positive integer n. In a density formula, its main role is normalization, meaning it allows the total area under the curve to be equal to 1.

1.6.6 Beta

If $X \sim \mathrm{Beta}(\alpha, \beta)$, then

$$f(x) = \frac{x^{\alpha-1}(1-x)^{\beta-1}}{B(\alpha, \beta)}, \qquad 0 < x < 1.$$

Because its possible values lie between 0 and 1, the beta distribution is especially useful for proportions and probabilities. Depending on α and β, it can be symmetric, skewed, concentrated near the middle, or concentrated near the endpoints.

Its mean is

$$E[X] = \frac{\alpha}{\alpha + \beta},$$

and its variance is

$$\operatorname{Var}(X) = \frac{\alpha\beta}{(\alpha+\beta)^2(\alpha+\beta+1)}.$$

The beta function is related to the gamma function by

$$B(\alpha,\beta) = \frac{\Gamma(\alpha)\Gamma(\beta)}{\Gamma(\alpha+\beta)}.$$

Later, beta distributions will provide convenient models for uncertainty about an unknown probability, such as the proportion of students who use a particular campus service.

1.6.7 Uniform

If $X \sim U(a,b)$, then

$$f(x) = \frac{1}{b-a}, \qquad a \le x \le b.$$

The uniform distribution assigns equal density everywhere within its interval. Its mean and variance are

$$E[X] = \frac{a+b}{2}, \qquad \operatorname{Var}(X) = \frac{(b-a)^2}{12}.$$

A uniform model might represent a random arrival time known only to lie within a fixed window, provided there is no reason to favor one part of the window over another.

```
par(mfrow = c(2, 2))

x_binom <- 0:15
plot(
  x_binom, dbinom(x_binom, size = 15, prob = 0.4),
  type = "h", lwd = 4, col = "steelblue",
  xlab = "Number of successes", ylab = "Probability",
  main = "Binomial(15, 0.4)"
)

x_poisson <- 0:15
plot(
  x_poisson, dpois(x_poisson, lambda = 4),
  type = "h", lwd = 4, col = "darkorange",
  xlab = "Number of events", ylab = "Probability",
```

```
  main = "Poisson(4)"
)

x_normal <- seq(-4, 4, length.out = 300)
plot(
  x_normal, dnorm(x_normal),
  type = "l", lwd = 2, col = "firebrick",
  xlab = "Value", ylab = "Density",
  main = "Normal(0, 1)"
)

x_beta <- seq(0, 1, length.out = 300)
plot(
  x_beta, dbeta(x_beta, shape1 = 2, shape2 = 5),
  type = "l", lwd = 2, col = "darkgreen",
  xlab = "Proportion", ylab = "Density",
  main = "Beta(2, 5)"
)

par(mfrow = c(1, 1))
```

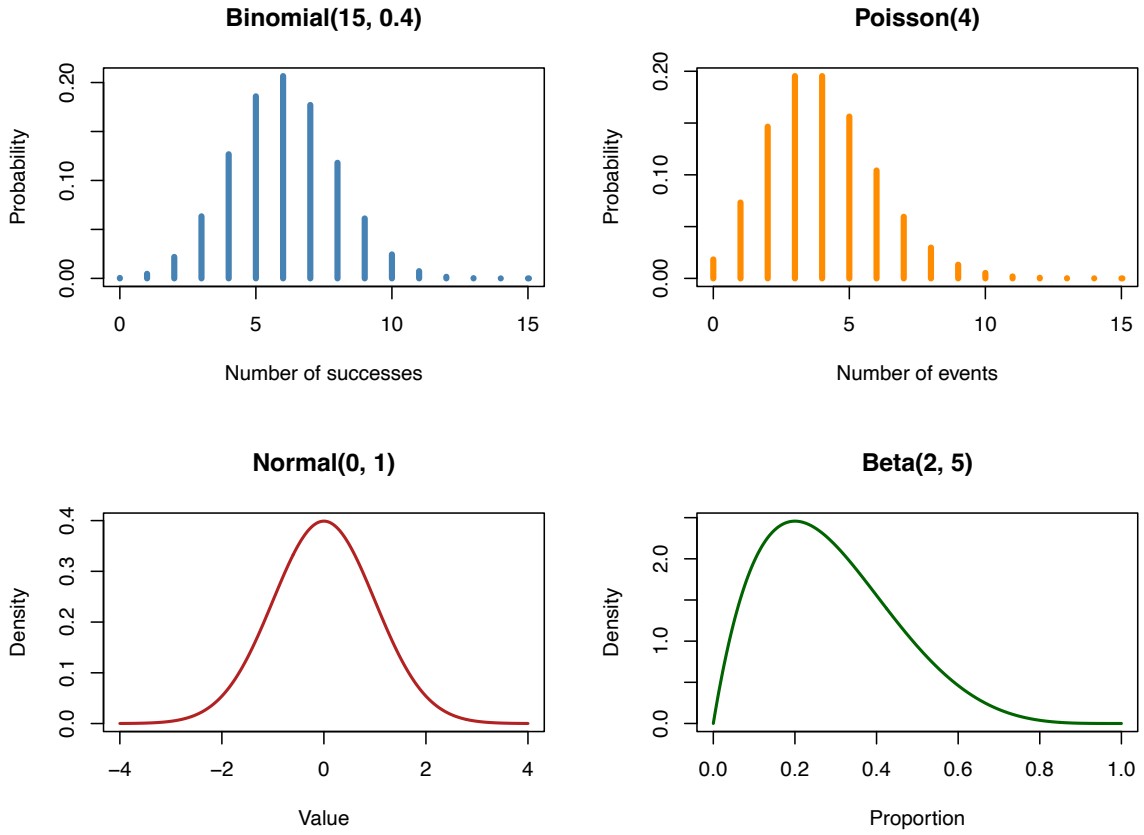


Figure 1.3: Several common distributions model distinctly different kinds of variation.

The plots in Figure 1.3 are not just different shapes. They live on different sets of possible values and answer different modeling needs. Binomial and Poisson variables are counts. Normal variables can take any real value. Beta variables are restricted to the interval from 0 to 1.

1.7 From individual observations to sample averages

Suppose $X_1, X_2, \ldots, X_n$ are observations from the same population. Their **sample mean** is

$$\bar{X} = \frac{1}{n} \sum_{i=1}^{n} X_i.$$

Notice that the sample mean is also a **random variable**. Two important probability results explain why sample means are so important in statistics. They answer different questions and

should not be confused.

1.7.1 The law of large numbers

The **law of large numbers** says that the sample mean $\bar{X}$ gets closer to the population mean $E[X]$ as the sample size grows.

This is a statement about **where the sample mean goes**. A small sample mean can be far from the population mean because of random variation. A large sample averages over more of that variation, making major departures less likely.

The running coin-toss proportion in Figure 1.1 is an example. Since a toss was coded 1 for heads and 0 for tails, its sample mean is exactly the sample proportion of heads. The population mean is 0.5, so the sample proportion settles near 0.5.

The law does not say that more data make every observation less variable. Individual coin tosses remain just as unpredictable. It says that the **average** becomes more stable as more and more data points are being included in that average.

1.7.2 The central limit theorem

The **central limit theorem** (CLT) says that the sampling distribution of the sample mean becomes approximately normal as n grows, even when the original population that generates the data is not normal.

This is a statement about **the shape of the distribution of sample means**. If the population distribution (the distribution that generates the data) has mean μ and finite standard deviation σ, then for a sufficiently large sample,

$$\bar{X} \text{ is approximately } N\left(\mu, \frac{\sigma^2}{n}\right).$$

Equivalently, the standard deviation of the sample mean is

$$\mathrm{SD}(\bar{X}) = \frac{\sigma}{\sqrt{n}}.$$

This formula explains why averages become less variable as the sample size grows.

To see the CLT, consider individual waiting times drawn from a strongly right-skewed exponential distribution. We repeatedly draw samples and save their means.

```r
set.seed(132)

one_wait <- rexp(5000, rate = 1)

sample_means <- replicate(
  5000,
  mean(rexp(30, rate = 1))
)

par(mfrow = c(1, 2))

hist(
  one_wait,
  breaks = 35,
  probability = TRUE,
  col = "skyblue",
  border = "white",
  main = "Individual waiting times",
  xlab = "Waiting time"
)

hist(
  sample_means,
  breaks = 30,
  probability = TRUE,
  col = "wheat",
  border = "white",
  main = "Means of 30 waiting times",
  xlab = "Sample mean"
)

curve(
  dnorm(x, mean = 1, sd = 1 / sqrt(30)),
  add = TRUE,
  col = "firebrick",
  lwd = 2
)

par(mfrow = c(1, 1))
```

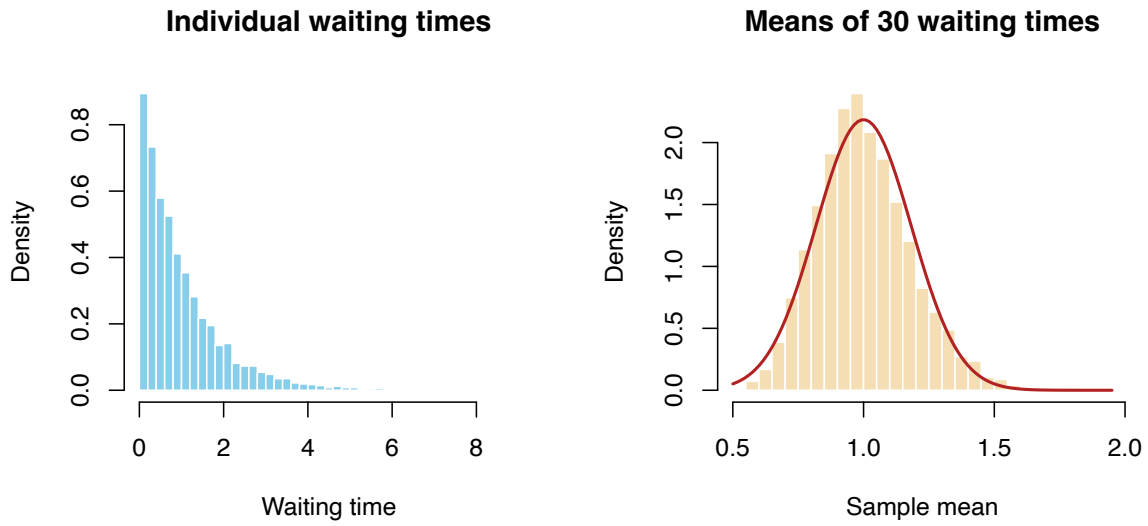


Figure 1.4: Individual waiting times are strongly skewed, but means of 30 waiting times have an approximately normal shape.

The left side of Figure 1.4 shows the population shape: most waits are short, but a few are quite long. The right side shows the distribution of averages from many samples of size 30. Those averages are more symmetric, concentrated near the population mean of 1, and closely matched by the normal curve.

The CLT does not say that the original population becomes normal, nor does it say that a large dataset must have a bell-shaped histogram. It describes the behavior of a statistic (specifically the sample mean) across repeated samples.

1.8 From probability to statistical inference

Probability and inference use the same mathematical model in opposite directions.

Probability starts with a model and asks about data. If a coin has probability $p = 0.5$ of heads, what is the probability of observing exactly 6 heads in 10 tosses? With $X \sim \text{Bin}(10, 0.5)$,

$$P(X = 6) = \binom{10}{6}(0.5)^6(0.5)^4 \approx 0.205.$$

In R:

```r
dbinom(6, size = 10, prob = 0.5)
```

```
[1] 0.2050781
```

Here the model and its parameter are known but the data are uncertain.

Statistical inference starts with data and asks about the model. If we toss a particular coin 10 times and observe 8 heads, what should we conclude about its unknown probability p of heads? Is $p = 0.5$ still plausible? What values of p are reasonably consistent with what we observed?

Here the data are known but the model or its parameters are uncertain. A natural initial estimate is $\hat{p} = 8/10 = 0.8$, but ten tosses contain limited information. Even a fair coin can occasionally produce eight or more heads:

```r
# Probability of at least 8 heads if the coin is fair
pbinom(7, size = 10, prob = 0.5, lower.tail = FALSE)
```

```
[1] 0.0546875
```

Statistical inference gives us tools to express that uncertainty rather than treating 0.8 as some unquestionable truth.

1.8.1 Population, sample, and model

The **population** is the full collection we want to understand. It might be all currently enrolled students at a university, every battery produced by a factory this week, or all future requests handled by a web service under stable conditions.

A **sample** is the subset we actually observe. We use the sample because measuring the entire population may be too costly, too slow, impossible, or conceptually inappropriate when the population includes future outcomes.

A **statistical model** is a set of probability distributions proposed as plausible descriptions of how the data were generated. The model connects the question about the population that we are interested in to the mathematics of probability. Every model simplifies reality in some way. The important question is whether its simplifications are reasonable for the purpose at hand.

1.9 Parametric and nonparametric approaches

A **parametric model** assumes that the population distribution belongs to a specified family described by a finite set of unknown parameters. For example,

$$X_i \sim N(\mu, \sigma^2)$$

assumes a normal population whose central tendency and spread are governed by two unknown numbers, μ and σ^2.

Parametric models can be efficient and interpretable because the model reduces a complicated distribution to a few quantities. That benefit depends on the assumptions being reasonable. A normal model may be useful for adult heights in a fairly homogeneous population, but it would be a poor model for personal income because income is bounded below and often strongly right-skewed.

A **nonparametric approach** makes fewer assumptions about the population's distributional form. Estimating a population median by using the sample median does not require us to assume that the population is normal, gamma, or any other particular shape.

"Nonparametric" does not mean "no parameters," and it does not mean "no assumptions." It means that we do not restrict the entire population distribution to a family indexed by a fixed, finite set of parameters. Stronger assumptions can give more precise conclusions when they are appropriate, while weaker assumptions provide flexibility and robustness.

1.10 Parameters, parameter spaces, and statistics

These three terms are closely related but refer to different things.

1.10.1 Parameters describe the population model

A **parameter** is an unknown numerical feature of a population or probability model. Examples include:

- the proportion p of students who commute by public transit
- the mean μ battery life for a model of laptop
- the variance σ^2 of delivery times
- the Poisson rate λ of help-desk requests per hour.

In this class, a parameter is fixed but unknown. It does not change from one random sample to another. However our information about it changes as we collect data.

The **parameter space** is the collection of values a parameter is allowed to take. The restrictions should match the meaning of the parameter. For a normal model $N(\mu, \sigma^2)$,

$$\mu \in \mathbb{R}, \qquad \sigma^2 > 0.$$

A mean can be any real number in the Normal probability model, while a variance must be positive. For a probability p, the parameter space is $[0, 1]$. Declaring the parameter space prevents nonsensical possibilities such as a probability of 1.4 or a negative variance.

1.10.2 Statistics are calculated from the sample

A **statistic** is any function of the observed data that does not depend on an unknown parameter:

$$T = t(X_1, X_2, \ldots, X_n).$$

Two essential statistics are the sample mean

$$\bar{X} = \frac{1}{n} \sum_{i=1}^{n} X_i$$

and the sample variance

$$S^2 = \frac{1}{n-1} \sum_{i=1}^{n} (X_i - \bar{X})^2.$$

The sample mean describes the center of the observed values. The sample variance describes their spread around that observed center.

Before data are collected, a statistic is also a random variable because its value depends on the random sample. After data are collected, we can calculate its realized value.

1.11 The main jobs of statistical inference

Most of this course can be organized around estimation, prediction, and hypothesis testing.

1.11.1 Point estimation

A **point estimate** is a single best guess for an unknown parameter. If 84 of 120 surveyed students support extending library hours, the sample proportion

$$\hat{p} = \frac{84}{120} = 0.70$$

is a natural point estimate of the population proportion p.

A point estimate is useful but incomplete. Another random sample would probably produce a slightly different value. Reporting only 0.70 can hide how much uncertainty remains.

1.11.2 Interval estimation

An **interval estimate** gives a range of parameter values supported by the data, together with a framework-specific measure of uncertainty. A confidence interval and a Bayesian credible interval may look similar numerically, but they attach probability to different objects and require different interpretations. We will develop both ideas later.

The essential motivation is the same though. An interval acknowledges that a sample cannot usually identify an unknown population quantity exactly.

1.11.3 Prediction

Prediction concerns an unobserved or future outcome rather than a fixed parameter. For example, enrollment data from the last ten years might be used to predict next year's STAT 132 enrollment.

Estimation and prediction are related but distinct. Estimating the average enrollment asks about a feature of the process. Predicting next year's enrollment asks about a new outcome, which can deviate from that average. A good prediction must account for both uncertainty about the model and natural year-to-year variation.

1.11.4 Hypothesis testing

A **hypothesis test** uses sample data to evaluate a claim about a parameter. Suppose a fruit distributor claims that the mean weight of a type of fruit is 100 grams. We might express the competing claims as

$$H_0 : \mu = 100 \qquad \text{and} \qquad H_1 : \mu \neq 100.$$

The null hypothesis H_0 is the specific claim being assessed. The alternative H_1 represents departures of interest. We ask whether the observed sample would be unusually inconsistent with H_0.

A test does not prove that one statement is true. Data contain randomness, so decisions can be wrong. A sample mean of 98 grams does not by itself refute a mean of 100: whether that difference is compelling depends on the sample size and the natural variability of fruit weights. Hypothesis testing provides a disciplined way to measure the evidence and control error rates.

1.12 How the pieces fit together

Statistical inference is possible because statistics have probability distributions. We begin with a population model containing unknown parameters, draw a random sample, and calculate a statistic. Probability theory describes how that statistic would vary across repeated samples. Inference then uses the statistic we actually observed to learn about the unknown parameter or predict a new observation.

The logic can be summarized as

$$\text{population model} \longrightarrow \text{random sample} \longrightarrow \text{statistic} \longrightarrow \text{inference about the population.}$$

The first arrows describe the data-generating direction studied by probability. The final arrow uses observed data to reason about what could have generated them.

1.13 Chapter summary

- A sample space lists possible outcomes, while events collect outcomes of interest.
- Conditional probability restricts attention to outcomes consistent with known information. Bayes' rule reverses the direction of conditioning while accounting for base rates.
- A random variable converts random outcomes into numerical values. Its distribution describes how probability is allocated.
- PMFs assign probability to discrete values, PDFs assign probability through area, and CDFs accumulate probability up to a value.
- Expectation describes a distribution's long-run center. Variance and standard deviation describe its spread.
- Common distributions represent different data types and mechanisms. Choosing one is a substantive modeling decision.

- The law of large numbers explains why sample averages stabilize. The central limit theorem explains why the distribution of many sample averages becomes approximately normal.
- Probability reasons from a model toward possible data. Statistical inference uses observed data to learn about an unknown model or parameter.
- Parameters describe populations or models. Statistics are computed from samples.
- Estimation, prediction, and hypothesis testing are three central goals of inference.